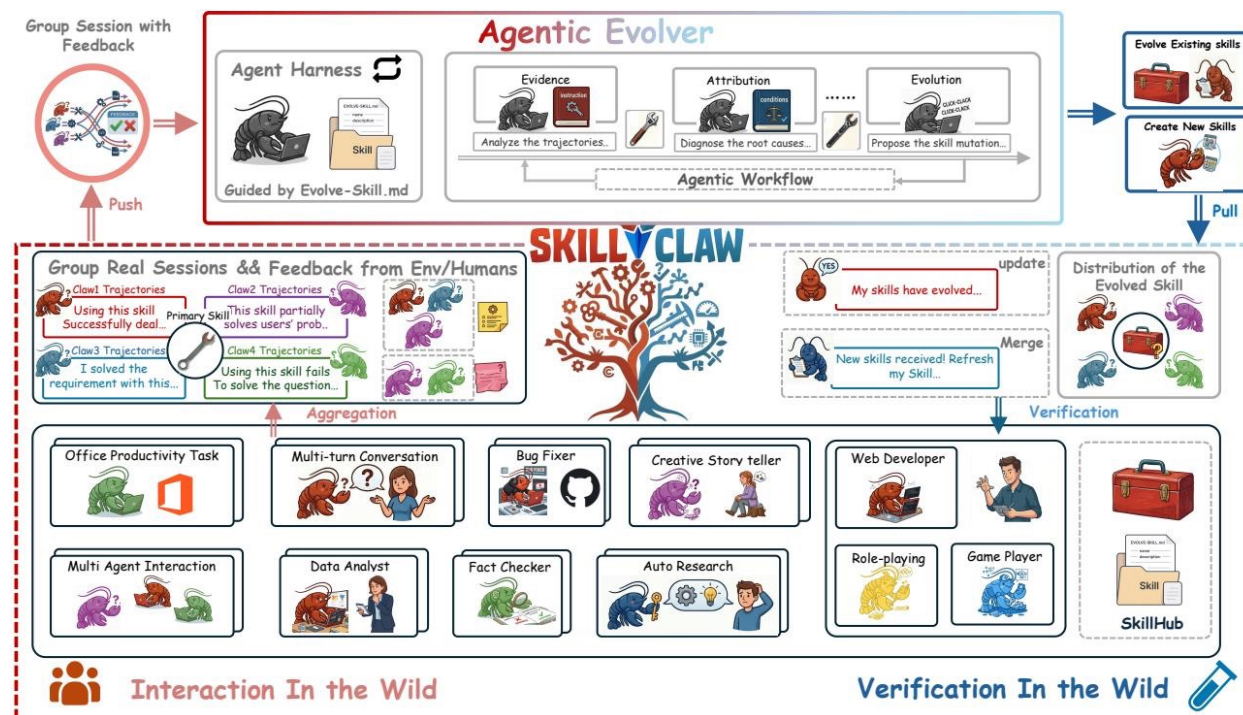


SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

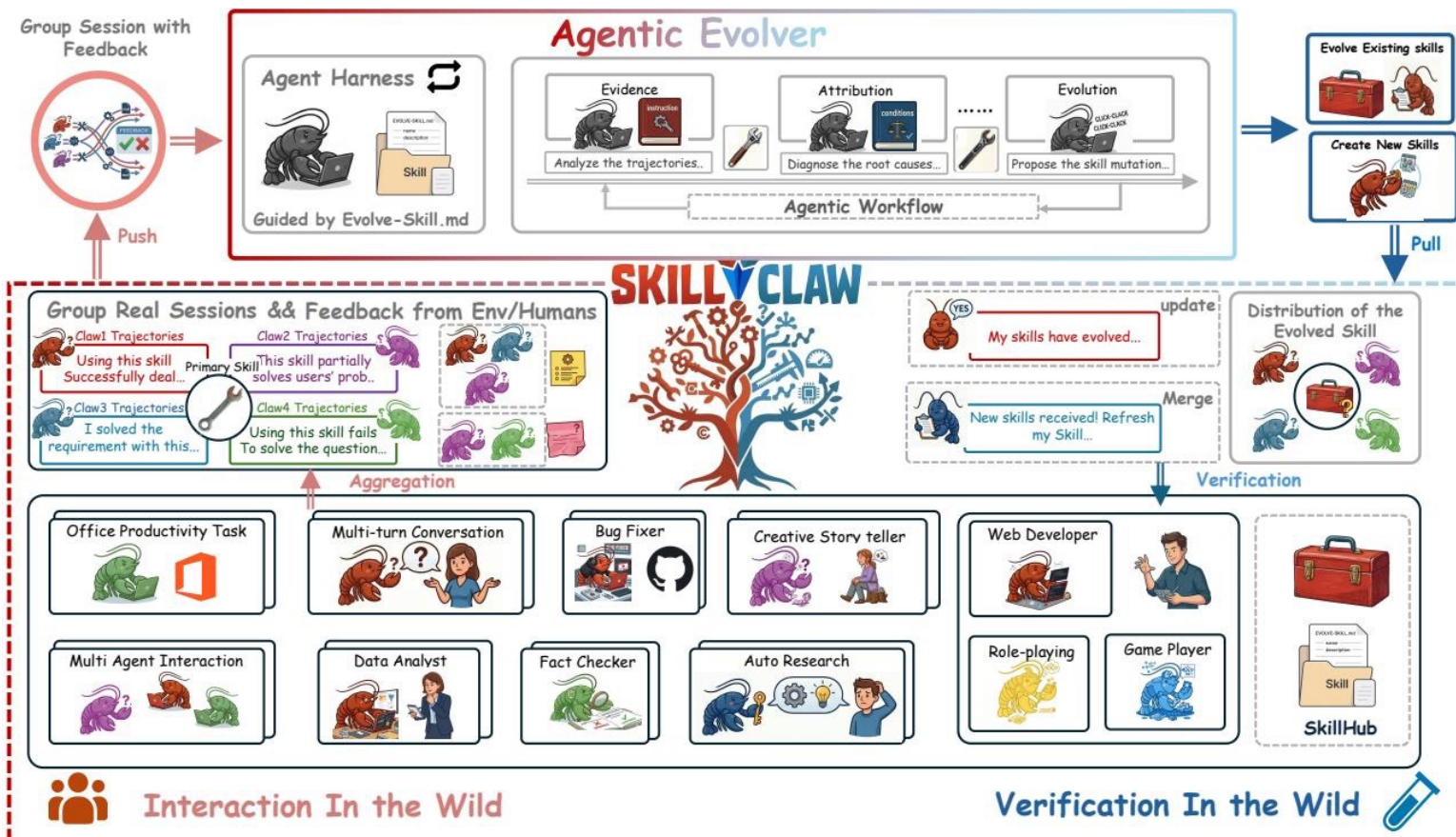
Source: arxiv.org/abs/2604.08377



Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

- Current LLM agents rely on reusable skills from centralized hubs, but these remain static after deployment.
- Similar tasks recur across different users, yet the same patterns of failure and recovery are repeatedly observed.
- Failures and solutions are rediscovered repeatedly across users, preventing system-level knowledge accumulation.
- Existing approaches fall short:
 - Memory-based methods store past trajectories but remain tied to specific instances and are hard to generalize.
 - Static skill libraries compress experience into structured instructions but do not evolve through usage.

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



The closed-loop pipeline consists of 4 main stages:

Evidence (Multi-user Interaction & Session Collection)

Attribution (Diagnosing root causes)

Evolution (Proposing skill mutations)

Distribution (Skill Synchronization)

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

- The agentic evolver analyzes each skill-specific group through three stages:
 - Evidence: Analyze interaction trajectories to identify behavioral patterns (both success and failure).
 - Attribution: Diagnose the root causes of failures and successes.
 - Evolution: Propose skill mutations—refining existing skills or creating new ones.
- Evolution is driven by open-ended agent reasoning:
 - Uses an autonomous agent to evaluate evidence rather than predefined update rules.
 - Produces context-aware skill updates that are robust across use cases.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

- Trajectories are aggregated across multiple users and diverse ecosystems:
 - Compatible with OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, and NemoClaw.
- Why cross-user aggregation is necessary:
 - Different users exercising the same skill under diverse contexts produce complementary signals.
 - Reveals both the conditions under which a skill works and those under which it breaks.
 - A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Highlight: Only 1 of 6 candidates was accepted, illustrating the conservative verification process that prevents regressions.

Day	Candidate Skill	Skill Function	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract from PDF/video/image)	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation	Reject	Keep current best pool
5	multimodal-creative-task-pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	Reject	Not admitted to next cycle

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Highlight: 3 of 6 days yielded accepted updates, showing successful iterative refinement before reaching a plateau.

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added "push hang treated as auth failure"; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	Reject	Not admitted to next cycle

Key Takeaways: Three Properties That Distinguish SkillClaw

- Collective Evolution
 - Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem.
- Fully Automatic
 - Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.
- Agentic Evolution Paradigm
 - Skill updates are produced through open-ended reasoning rather than predefined update rules.
- Implications for AI Systems Engineering:
 - Paves the way for building continuously self-improving agent systems.